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Seminar time and location: Mon, 12:00-13:45; C5 3, Room 0.23
Consultation hours: Tuesday 14:00 – 16:00

Research Seminar

Quantitative Research in Political Science

"How to Apply the Classical Toolbox"
&
"How to Analyse Digital Content for Political Science Research"

Document version: 17.04.2025

General information

Module: The aim of the entire module is to prepare students for their Bachelor's thesis. In this respect, students will:

- learn to critically discuss current research contributions,
- deepen their methodological skills by critically evaluating current research literature,
- learn to develop a research question and draft a suitable research design;
- practise how to carry out and write up their own empirical work in an exemplary manner.

Aim: The project seminar aims to help students in (i) developing their own questions on European politics research, and (ii) a suitable design for addressing said questions, (iii) analysing relevant data with proper methods, (iv) effectively communicating their results, and (v) drawing substantive conclusions from their empirical analyses.

Seminar: The seminar provides students with a comprehensive overview of the main quantitative tools used in empirical political science and aims to enhance their ability to develop scientifically rigorous research designs. The course will begin by enhancing students' knowledge of the R programming language and classical statistical methods. The course then will move to methods for the quantitative analysis of text. Students will be guided in developing their own research designs, with particular emphasis on translating concepts into measures, moving from research questions to empirically testable hypotheses, and selecting appropriate analytical tools to examine their data.

Related seminar: This seminar is closely linked with the seminar taught by Prof. Dr. Daniela Braun's 'Current Developments in Political Science Research on Europe: "Legitimacy Problems and Polarisation in European Societies"'.[2] In particular, students will replicate a selection of studies addressed in the above mentioned seminar.

Literature: We will only make the literature used in the seminar available to you via Teams if

we don't have access to it through the UdS libraries. Since the library now offers sufficient access to political science journals and books, I assume that you have researched the literature. In case of any problems accessing specific texts, please let us know as soon as possible (i.e. not the night before the deadline).

Consultation: In case of consultation/coordination, you are welcome to come to our office hours after making an appointment via email.

Examination: The examination for this course is a graded term paper, based on the presentation prepared for the course taught by Prof. Dr. Braun Current Developments in Political Science Research on Europe: "Legitimacy Problems and Polarisation in European Societies"'.

The term paper should include the following elements:[1] [^2]

1. Introduction: Why is the theme exciting and important? What is the social and scientific relevance?
2. Research question(s): What is the specific question addressed by the paper?
3. State of the art: What knowledge do we have already about this topic and/or research questions?
4. Research design: How are the research questions going to be investigated?
    a. Concepts – i.e., What's your analytical model?
    b. Case selection – i.e., Country selection, time period analysed. Why is the case relevant for your research questions?
    c. Data – i.e., Why are the chosen data useful for addressing your research questions? What are the main characteristics of the data?
    d. Operationalization/Measurements– i.e., which variables and how are you going to manipulate them?
    e. Methods – i.e., what's your statistical model?

5. Results: What are the estimates of your models and how they can be interpreted? Do your results align with your expectations or not?
6. Discussion of the results and conclusions: What are the implications of the results for the topic under investigation?

Session overview

Pt. I - How to Apply the Classical Toolbox

Class 1 [07.04] – Intro: Motivation, materials, aims, requirements

Content:

Presentation of seminar rules and the two-part structure of the course (first classical quantitative methods in R, then digital content analysis). Overview of the curriculum, course objectives. Core concepts such as models, populations vs. samples, observations/variables, and correlation versus causation are defined. Three assignment: refreshing R skills, completing key readings, and beginning to specify a research model

[1] See also: Gschwend, Thomas & Schimmelfennig, Frank (2007): Introduction: Designing research in political science - a dialogue between theory and data. In: Gschwend, Thomas & Frank Schimmelfennig (eds.): Research design in political science: how to practice what they preach (pp. 1-18). Houndmills: Palgrave Macmillan. https://nbn-resolving.org/urn:nbn:de:0168-ssoar-258302.

(in particular, the dependent variable).

Literature:

- Bueno de Mesquita & Fowler (2021), Chapters 2,3,4
- King (1998) Chapter 1

Other sources:

- A platform to practice R without installation: https://adp-cvk.quarto.pub/r-for-social-science-data-analysis/
- Some basic statistics with R: https://statsandr.com/blog/descriptive-statistics-in-r/

Class 2 [14.04] – The first bricks: Data structures, statistical concepts, Git R basics

Content:

Introduced R projects, folder structures, basic R objects and data structures, and coding principles such as KIS, DRY, comments, and clear scripts. Revisited populations, samples, observations, variables, and correlation, then covered central tendency, dispersion, skewness, outliers, and the Central Limit Theorem. The practical session focused on setting up Git, GitHub, GitHub Desktop, and a local R project/repository.

Literature:

- Agresti & Finlay (2009), Chapters 1, 2, 3

Other sources:

- Git and Github for beginners: https://product.hubspot.com/blog/git-and-github-tutorial-for-beginners

Class 3 [28.04] – Going multivariate! Generalised linear models, and the bread and butter of multivariate methods (i.e., the linear model)

Content:

Covered descriptive statistics for categorical and continuous variables, including dispersion, quantiles, and the appropriate interpretation of different measures. Introduced statistical inference through the Central Limit Theorem, confidence intervals, null and alternative hypotheses, p-values, and z- and t-tests. The R component introduced tidyverse workflows and the 2019 European Election Study (EES), including data loading, variable selection, distribution checks, and missing values.

Literature:

- Agresti & Finlay (2009), Chapters 4, 5, 6, 7

Other sources

- Wickham et al. (2019), "Welcome to the tidyverse": https://doi.org/10.21105/joss.01686
- Course GitHub repository and workflow handout.

Class 4 [05.05] – Correlation, survey data, and first analyses in R

Content:

Introduced association and linear correlation, its interpretation from −1 to +1, and

covariance as the non-standardised measure underlying correlation and variance–covariance matrices. Applied these concepts to the 2019 EES voter study, discussing the scope and limits of cross-sectional survey data. The practical session covered tidy data, loading and merging data, filtering by country, descriptive comparisons, and correlation plots.

Literature:

- Agresti & Finlay (2009), Chapters 1, 2, 3

Other sources:

- 2019 European Election Study voter survey
- Course GitHub repository
- RStudio shortcuts: https://docs.posit.co/ide/user/ide/reference/shortcuts.html

Class 5 [05.05] – Generalised Linear Models and quantities of interest

Content:

Introduced Generalised Linear Models as a unified framework consisting of an outcome distribution, linear predictor, and link function, with the linear model as a special case. Revisited the interpretation of regression coefficients and introduced predicted values, marginal effects, and average marginal effects as substantively interpretable quantities of interest. The practical session applied linear models to EES data and examined how predictions change across covariate values.

Literature:

- Gill (2001)
- Gill & Torres (2020)

Other sources:

- Heiss, A. (2022, May 20). *Marginalia: A guide to figuring out what the heck marginal effects, marginal slopes, average marginal effects, marginal effects at the mean, and all these other marginal things are*. https://doi.org/10.59350/40xaj-4e562

Class 6 [12.05] – From OLS to logistic regression

Extended the GLM framework to binary outcomes, contrasting the linear probability model with logistic regression. Discussed why linear models can produce invalid probabilities and violate key assumptions with binary dependent variables, and how the logit model addresses these issues. Applied predicted probabilities and average marginal effects to logit models, emphasising that both can depend on the values of other covariates.

Literature:

- Gill (2001)
- Gill & Torres (2020)

Other sources:

- Heiss, A. (2022, May 20). *Marginalia: A guide to figuring out what the heck marginal effects, marginal slopes, average marginal effects, marginal effects at the mean, and all these other marginal things are*. https://doi.org/10.59350/40xaj-4e562

Pt. II - How to Analyse Digital Content for Political Science Research

The articles highlighted in blue are required reading for the course. The other articles are also useful for each week's theme, but select the ones which are most helpful for your coursework.

Class 7 [26.05] – Introduction: course details, technical set up, where to find content

Content: In this class, we will introduce the basics of how to use R to analyse text as data. We will start with an overview of the type of text we can work with and the type of concepts we can measure. We will review practical skills such as importing and preparing the data, as well as learning key terms for future use. For this, we will learn how to use the R package 'Quanteda' and its key functions. In particular, we will learn how to use functions to create simple descriptive statistics from a group of documents and other means of summarising their contents.

Literature: Grimmer, J. and Stewart, B.M. (2013) 'Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts', *Political Analysis*, 21(3), pp. 267–297. doi:10.1093/pan/mps028. (see Teams group for a PDF)

Other sources:

A selection of useful political texts can be found here:

Parlspeech: https://search.gesis.org/research_data/SDN-10.7802-2824

Party Manifestos: https://manifesto-project.wzb.eu/datasets/MPDS2024a

Further examples: https://www.ics.uci.edu/~smyth/courses/cs175/text_data_sets.html

Class 8 [02.06] – Getting an Overview: using dictionaries to measure quantities at scale

Content: In this class, we will learn the first important method for quantifying text: Dictionaries. We will practice preparing the text, importing the dictionary, and using it to measure key concepts within the text. We will discuss the pros and cons of this approach, and consider other concepts which it can be used for. Finally, we will create our own dictionaries and relate them to the topics of interest for our coursework, including polarisation and political trust.

Literature:

Pauwels (2011) Measuring Populism: A Quantitative Text Analysis of Party Literature in Belgium. Journal of Elections, Public Opinion and Parties 21(1): 97-119.

Schmader et al (2007) A Linguistic Comparison of Letters of Recommendation for Male and Female Chemistry and Biochemistry Job Applicants. Sex roles, 57(7-8): 509–514.

Young, L., & Soroka, S. (2012). Affective news: The automated coding of sentiment in political texts. Political communication, 29(2), 205-231.

Simonsen, K.B. and Widmann, T. (2025) 'When Do Political Parties Moralize?: A Cross-National Study of the Use of Moral Language in Political Communication on Immigration', *British Journal of Political Science*, 55, p. e33. https://doi:10.1017/S0007123425000122

Other sources:

You can find other useful dictionaries and related information here:

https://provalisresearch.com/products/content-analysis-software/wordstat-dictionary/

http://gender-decoder.katmatfield.com/about#masculine

https://github.com/quanteda/quanteda.sentiment

https://github.com/kbenoit/quanteda.dictionaries/tree/master/data

Class 9 [16.06] – Operationalising results: text analysis to measure polarisation

Content: In this class, we will distinguish between different types of political polarisation (elite vs mass, ideological vs affective) and consider how to use text analysis to measure these concepts, with reference to the overview by Bauer (2019). We will apply some of the polarisation measures using the text analysis methods we have learned so far. We will also get an overview of the final text preparation steps and discuss how and when they can be useful.

Literature:

Bauer, P. C. (2019). Conceptualizing and measuring polarization: A review. https://doi.org/10.31235/osf.io/e5vp8

Heyne, L., Navarrete, R. M., & Katsanidou, A. (2025). Fading EUphoria: party system polarisation and EU attitudes in Greece, Portugal and Spain. *South European Society and Politics*, 1–29. https://doi.org/10.1080/13608746.2025.2468930

Muñoz, P., Bellogín, A., Barba-Rojas, R. *et al.* Quantifying polarization in online political discourse. *EPJ Data Sci.* **13**, 39 (2024). https://doi.org/10.1140/epjds/s13688-024-00480-3

Koedam, J., Binding, G. and Steenbergen, M.R. (2025) 'Multidimensional Party Polarization in Europe: Cross-Cutting Divides and Effective Dimensionality', *British Journal of Political Science*, 55, p. e24. https://doi:10.1017/S0007123424000474

Kubin, E., & von Sikorski, C. (2021). The role of (social) media in political polarization: a systematic review. *Annals of the International Communication Association*, *45*(3), 188–206. https://doi.org/10.1080/23808985.2021.1976070

Schäfer, C., Popa, S. A., Braun, D., & Schmitt, H. (2020). The reshaping of political conflict over Europe: from pre-Maastricht to post-'Euro crisis.' *West European Politics*, *44*(3), 531–557. https://doi.org/10.1080/01402382.2019.1709754

Other sources: tbd

Class 10 [23.06] – Going Deeper I: scaling and classifying with machine learning

Content: In this class, we will learn how to use the Quanteda objects to develop models for classification. Each model uses the patterns from our dfms (text represented as numbers) to group together similar documents. This enables us to distinguish for example between texts which have a positive and negative tone, a pro or anti stance, or a left or right wing position. Again, the results can be used to measure political concepts such as polarisation for inclusion in statistical analysis such as regression models. As we the earlier dictionary methods, we will also discuss the advantages and disadvantages of these techniques.

Literature: Cowburn, M. and Sältzer, M. (2025) 'Partisan communication in two-stage elections: the effect of primaries on intra-campaign positional shifts in congressional elections', *Political Science Research and Methods*, 13(2), pp. 392–411. https://doi:10.1017/psrm.2023.62

Schafer, D. (2023). Sycophants in 280 characters: Using Twitter to measure the authoritarian sentiment of presidential advisors in Turkey. *Party Politics*, *30*(2), 379-386. https://doi.org/10.1177/13540688231153099 (Original work published 2024)

Oliveira, D. J. S., Bermejo, P. H. de S., & dos Santos, P. A. (2016). Can social media reveal

the preferences of voters? A comparison between sentiment analysis and traditional opinion polls. *Journal of Information Technology & Politics*, *14*(1), 34–45. https://doi.org/10.1080/19331681.2016.1214094

Shah, D. V., Hanna, A., Bucy, E. P., Wells, C., & Quevedo, V. (2015). The Power of Television Images in a Social Media Age: Linking Biobehavioral and Computational Approaches via the Second Screen. *The ANNALS of the American Academy of Political and Social Science*, *659*(1), 225-245. https://doi.org/10.1177/0002716215569220 (Original work published 2015)

Other sources: tbd

Class 11 [30.06] – Going Deeper II: scaling and classifying with word embeddings

Content: In this class, we will study a more recent advance in text classification, word embeddings. This both an useful classification method by itself, and also an important building block for later technological developments, including large language models (LLMs) like ChatGPT. We will learn the basis and code for calculating word embeddings and apply some of the foundational embedding models such as GloVe and Word2Vec to our texts. Then we will link the results with the measurements learned in Week 9 to create calculations of political polarisation in European countries.

Literature: Rheault, L. and Cochrane, C. (2020) 'Word Embeddings for the Analysis of Ideological Placement in Parliamentary Corpora', *Political Analysis*, 28(1), pp. 112–133. doi:10.1017/pan.2019.26.

Martin Haselmayer, Sarah C Dingler, Marcelo Jenny, How Women Shape Negativity in Parliamentary Speeches—A Sentiment Analysis of Debates in the Austrian Parliament, *Parliamentary Affairs*, Volume 75, Issue 4, October 2022, Pages 867–886, https://doi.org/10.1093/pa/gsab045

Cochrane, C., Rheault, L., Godbout, J. F., Whyte, T., Wong, M. W. C., & Borwein, S. (2021). The Automatic Analysis of Emotion in Political Speech Based on Transcripts. *Political Communication*, *39*(1), 98–121. https://doi.org/10.1080/10584609.2021.1952497

Rozado, D., & Al-Gharbi, M. (2022). Using word embeddings to probe sentiment associations of politically loaded terms in news and opinion articles from news media outlets. *Journal of Computational Social Science*, *5*(1), 427-448.

Other sources: tbd

Class 12 [07.07] – Review, New Developments: analysing text with Large Language Models Other content: images, video, and the future of online content analysis

Content: In our final class before the presentations, we will bring our methods up to date by studying the use of large language models in text classification and recent advances in the classification of other online content such as images and video. We will learn about the variety of models available, their costs and benefits, and ways of determining their accuracy while balancing other potential drawbacks of using them for our work. We will also discuss additional ways to access useful data from online sources, including web scraping methods and current legal questions for gaining access to social media data. Finally, we will take time to review what we have learned so far and address any questions about previous classes.

Literature: Widmann, T. and Simonsen, K.B. (2025) 'Setting the tone: the diffusion of moral and moral-emotional appeals across political and public discourse', *Political Science*

Research and Methods, 13(2), pp. 489–496. doi:10.1017/psrm.2024.34.

Widmann, T., & Wich, M. (2023). Creating and comparing dictionary, word embedding, and transformer-based models to measure discrete emotions in German political text. *Political Analysis*, *31*(4), 626-641.

Burnham, M. (2024) 'Stance detection: a practical guide to classifying political beliefs in text', *Political Science Research and Methods*, pp. 1–18. doi:10.1017/psrm.2024.35.

Ornstein, J.T., Blasingame, E.N. and Truscott, J.S. (2025) 'How to train your stochastic parrot: large language models for political texts', *Political Science Research and Methods*, 13(2), pp. 264–281. https://doi:10.1017/psrm.2024.64

Ivanusch, C. (2024). Where do parties talk about what? Party issue salience across communication channels. *West European Politics*, *48*(3), 618–644. https://doi.org/10.1080/01402382.2024.2322234

Daniel, W. T., & Obholzer, L. (2025). Social media campaigning across multiple platforms: evidence from the 2024 European elections. *West European Politics*, 1–27. https://doi.org/10.1080/01402382.2025.2501464

Tarr, A., Hwang, J. and Imai, K. (2023) 'Automated Coding of Political Campaign Advertisement Videos: An Empirical Validation Study', *Political Analysis*, 31(4), pp. 554–574. Tarr, A., Hwang, J. and Imai, K. (2023) 'Automated Coding of Political Campaign Advertisement Videos: An Empirical Validation Study', *Political Analysis*, 31(4), pp. 554–574. doi:10.1017/pan.2022.26.doi:10.1017/pan.2022.26.

Other sources: https://researchmethodscommunity.sagepub.com/blog/image-as-data-automated-visual-content-analysis-for-social-science

Class 13 [14.07] – Presentation of Research Plans and Initial Findings

Content: tbd

Literature: tbd

Other sources: tbd

References

Required readings

Agresti, A., & Finlay, B. (2014). *Statistical methods for the social sciences* (4. ed., Pearson new internat. ed). Pearson.

Bueno de Mesquita, E., & Fowler, A. (2021). *Thinking clearly with data: A guide to quantitative reasoning and analysis*. Princeton University Press.

King, G. (1998). *Unifying Political Methodology: The Likelihood Theory of Statistical Inference*. University of Michigan Press.

Optional readings

Brambor, T., Clark, W. R., & Golder, M. (2005). Understanding interaction models: improving empirical analyses. *Political Analysis*, *14*(1), 63–82. https://doi.org/10.1093/pan/mpi014

Cinelli, C., Forney, A., & Pearl, J. (2022). A crash course in good and bad controls. *Sociological Methods & Research*, 53(3), 1071–1104. https://doi.org/10.1177/00491241221099552

Gill, J. (2001). *Generalized Linear Models*. SAGE Publications, Inc.

Gill, J., & Torres, M. (2020). *Generalized linear models: a unified approach - second edition*. (Vols. 1-0). SAGE Publications, Inc., https://methods.sagepub.com/book/mono/generalized-linear-models-2e/toc

Gschwend, T., & Schimmelfennig, F. (Eds.). (2007). *Research design in political science: How to practice what they preach*. Palgrave Macmillan.

Heiss, A. (2022, May 20). Marginalia: A guide to figuring out what the heck marginal effects, marginal slopes, average marginal effects, marginal effects at the mean, and all these other marginal things are. *andrewheiss.com*. https://doi.org/10.59350/40xaj-4e562

King, G. (1998). *Unifying political methodology*. https://doi.org/10.3998/mpub.23784

Long, J. S. (1997). *Regression models for categorical and limited dependent variables*. SAGE.

Lundberg, I., Johnson, R., & Stewart, B. M. (2021). What is your estimand? Defining the target quantity connects statistical evidence to theory. *American Sociological Review*, *86*(3), 532–565. https://doi.org/10.1177/00031224211004187